CEZERİ BİLİM VE TEKNOLOJİ OKULLARI

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2026 – 2027 Academic Year

First Quarter Examination

Mark scheme — FOR TEACHER USE ONLY

Class:	11 / _____	**Name:**	_____
Course:	Physics — Grade 11 (AS Level)	**Date:**	_____
Version:	A	**Duration:**	75 minutes

Marking codes: M = method mark, A = accuracy mark (depends on the M mark), B = independent mark, C = compensatory (award if the final answer is correct). Allow error carried forward (ECF) from an earlier part. Accept answers within ±1 in the last significant figure, and work done with $k = 9.0 \times 10^9$ N m² C⁻².

Structure

Section	Style	Questions	Marks
A — Multiple choice	Paper 1 style	1–15	15 (30%)
B — Structured questions	Paper 2 style	16–19	25 (50%)
C — Practical skills	Paper 3 style	20	10 (20%)

Weighting

Lesson	Questions	Count	Marks
Speed, velocity and acceleration	4, 5, 6, 7, 17, 20	6	20
Distance and displacement	2, 3, 16	3	8
Vectors: resolving and combining	1, 8, 9, 10, 11, 18	6	11
Coulomb's law (2 and 3 charges; line and right-angled triangle)	12, 13, 14, 15, 19	5	11

Section A answer key

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	D	A	B	A	B	C	C	A	C	B	D	B	D	A	C

Question 1 *[1 mark] — Vectors*

Part	Answer / marking points	Marks
Answer D	Displacement has both magnitude and direction; distance, speed and time are scalars Correct option: D B1	1

Question 2 *[1 mark] — Distance and displacement*

Part	Answer / marking points	Marks
Answer A	Average speed = 400 ÷ 80 = 5.0 m s^{-1} The athlete finishes at the start, so displacement = 0 and average velocity = 0 Correct option: A B1	1

Question 3 *[1 mark] — Distance and displacement*

Part	Answer / marking points	Marks
Answer B	Distance = 400 + 300 = 700 m Displacement = 400 − 300 = 100 m east Correct option: B B1	1

Question 4 *[1 mark] — Speed, velocity and acceleration*

Part	Answer / marking points	Marks
Answer A	Velocity = gradient of section R = (0 − 12) ÷ (10 − 7) = −4.0 m s^{-1} Correct option: A B1	1

Question 5 *[1 mark] — Speed, velocity and acceleration*

Part	Answer / marking points	Marks
Answer B	a = (25 − 5.0) ÷ 8.0 = 2.5 m s^{-2} Correct option: B B1	1

Question 6 *[1 mark] — Speed, velocity and acceleration*

Part	Answer / marking points	Marks
Answer C	Constant acceleration gives a straight, sloping velocity–time line Correct option: C B1	1

Question 7 *[1 mark] — Speed, velocity and acceleration*

Part	Answer / marking points	Marks
Answer C	Average speed = total distance ÷ total time = 60 ÷ 1.5 = 40 km h^{-1} (not the mean of 60 and 30) Correct option: C B1	1

Question 8 *[1 mark] — Vectors*

Part	Answer / marking points	Marks
Answer A	θ is measured from the vertical, so the vertical component is F cos θ and the horizontal component is F sin θ Correct option: A B1	1

Question 9 *[1 mark] — Vectors*

Part	Answer / marking points	Marks
Answer C	$R = \sqrt{(5.0^2 + 8.0^2)} = \sqrt{89} = 9.4$ N Correct option: C B1	1

Question 10 *[1 mark] — Vectors*

Part	Answer / marking points	Marks
Answer B	horizontal component $= \sqrt{(15^2 - 9.0^2)} = \sqrt{144} = 12$ m s^{-1} Correct option: B B1	1

Question 11 *[1 mark] — Vectors*

Part	Answer / marking points	Marks
Answer D	The resultant lies between 12 − 5.0 = 7.0 N and 12 + 5.0 = 17 N, so 19 N is impossible Correct option: D B1	1

Question 12 *[1 mark] — Coulomb's law*

Part	Answer / marking points	Marks
Answer B	$F \propto Q_1Q_2 \div r^2$: × 2 for the charge and ÷ 2^2 = ÷ 4 for the distance, giving F/2 Correct option: B B1	1

Question 13 *[1 mark] — Coulomb's law*

Part	Answer / marking points	Marks
Answer D	$F = 8.99 \times 10^9 \times 2.0 \times 10^{-6} \times 3.0 \times 10^{-6} \div 0.30^2 = 0.60$ N Unlike charges attract Correct option: D B1	1

Question 14 *[1 mark] — Coulomb's law*

Part	Answer / marking points	Marks
Answer A	The two charges are equal and the same distance d from M, so the two forces on +q are equal and opposite: resultant = 0 Correct option: A B1	1

Question 15 *[1 mark] — Coulomb's law*

Part	Answer / marking points	Marks
Answer C	The two forces of size F are at right angles: $\sqrt{(F^2 + F^2)} = \sqrt{2}\,F$ Correct option: C B1	1

Question 16 *[6 marks] — Distance and displacement; speed and velocity*

Part	Answer / marking points	Marks
(a)	Distance is a scalar (magnitude only, total path length); displacement is a vector (has direction; straight line from start to finish) B1	1
(b)	600 + 800 = 1400 m A1	1
(c)	displacement = $\sqrt{(600^2 + 800^2)} = 1000$ m A1 angle = $\tan^{-1}(800 \div 600) = 53°$ north of east B1	2
(d)(i)	total time = 120 + 200 = 320 s; average speed = 1400 ÷ 320 = 4.4 m s^{-1} A1	1
(d)(ii)	average velocity = 1000 ÷ 320 = 3.1 m s^{-1} A1 (allow ECF from (c))	1

Question 17 *[6 marks] — Speed, velocity and acceleration*

Part	Answer / marking points	Marks
(a)	$v = u + at = 0 + 2.5 \times 6.0 = 15$ m s^{-1} A1	1
(b)	$s = \frac{1}{2}at^2 = \frac{1}{2} \times 2.5 \times 6.0^2 = 45$ m A1	1
(c)	$s = 15 \times 20 = 300$ m A1 (allow ECF from (a))	1
(d)	$a = (0 - 15) \div 5.0 = -3.0$ m s^{-2} A1 (allow "deceleration of 3.0 m s^{-2}")	1
(e)	total distance = 45 + 300 + 37.5 = 382.5 m and total time = 6.0 + 20 + 5.0 = 31 s M1 average speed = 382.5 ÷ 31 = 12 m s^{-1} A1 (allow 12.3)	2

Question 18 *[6 marks] — Vectors: resolving and combining*

Part	Answer / marking points	Marks
(a)	A scalar has magnitude only; a vector has magnitude and direction B1	1
(b)(i)	$50 \cos 35° = 41$ N A1	1
(b)(ii)	$50 \sin 35° = 29$ N A1	1
(c)	$v = \sqrt{(15^2 + 6.0^2)}$ M1 v = 16 m s^{-1} A1 (allow 16.2) $\theta = \tan^{-1}(6.0 \div 15) = 22°$ above the horizontal B1	3

Question 19 *[7 marks] — Coulomb's law: charges in a line and on a right-angled triangle*

Part	Answer / marking points	Marks
(a)	Attractive, because B and C have opposite (unlike) signs B1	1

Part	Answer / marking points	Marks
(b)(i)	$F = 8.99 \times 10^{9} \times 3.0 \times 10^{-6} \times 2.0 \times 10^{-6} \div 0.10^{2} = 5.4$ N A1	1
(b)(ii)	$F = 8.99 \times 10^{9} \times 2.0 \times 10^{-6} \times 4.0 \times 10^{-6} \div 0.20^{2} = 1.8$ N A1	1
(c)	A repels B and C attracts B, so both forces act towards C: 5.4 + 1.8 = 7.2 N towards C (to the right) B1 Magnitude and direction both needed; 3.6 N (subtracting) scores 0	1
(d)(i)	$F(X) = 8.99 \times 10^{9} \times 3.0 \times 10^{-6} \times 2.0 \times 10^{-6} \div 0.30^{2} = 0.60$ N and $F(Y) = 8.99 \times 10^{9} \times 3.0 \times 10^{-6} \times 2.0 \times 10^{-6} \div 0.40^{2} = 0.34$ N B1 (both values needed)	1
(d)(ii)	resultant $= \sqrt{(0.60^{2} + 0.34^{2})} = 0.69$ N A1 angle $= \tan^{-1}(0.60 \div 0.34) = 61°$ (below OY) A1 (allow 60°–61°; ECF from (i))	2

Question 20 *[10 marks] — Practical skills: speed of a toy car*

Part	Answer / marking points	Marks
(a)(i)	metre rule (or tape measure / ruler) B1	1
(a)(ii)	stopwatch (or stop clock / timer) B1	1
(b)	d = 1.60 m: $1.60 \div 4.0 = 0.40$ m s^{-1} B1 d = 2.00 m: $2.00 \div 5.0 = 0.40$ m s^{-1} B1	2
(c)	all five points plotted correctly (within half a small square) B1 one straight, thin line of best fit through (or very close to) the origin B1	2
(d)	1.0 m (accept 0.95 – 1.05 m, read from the student's own line) A1	1
(e)	gradient $= 2.00 \div 5.0 = 0.40$ m s^{-1} (accept 0.38 – 0.42) A1	1
(f)	The speed of the car is constant (it does not change) B1	1
(g)	Any one: repeat each time measurement and take the mean; use light gates / a data logger; film the car and use the video timer B1 "Be more careful" scores 0	1

Model answer for Question 20(c) and (d)

